

0. 参考文档

参考的文档是这个：[驱动（RK3588S）第三课时：文件系统（rootfs）_rk3588 rootfs-CSDN 博客](#)，我下载的是 2025 年 11 月的源码。为了方便编译，在解压好的 buildroot 目录下新建一个脚本，然后 `chmod +x mk.sh`。内容如下图：

```
dev@localhost:~/self_linux/buildroot-2025.11$ cat mk.sh
#!/bin/bash

make ARCH=arm CROSS_COMPILE=aarch64-none-linux-gnu- $1 $2 $3 $4 $5 $6 $7 $8

dev@localhost:~/self_linux/buildroot-2025.11$
```

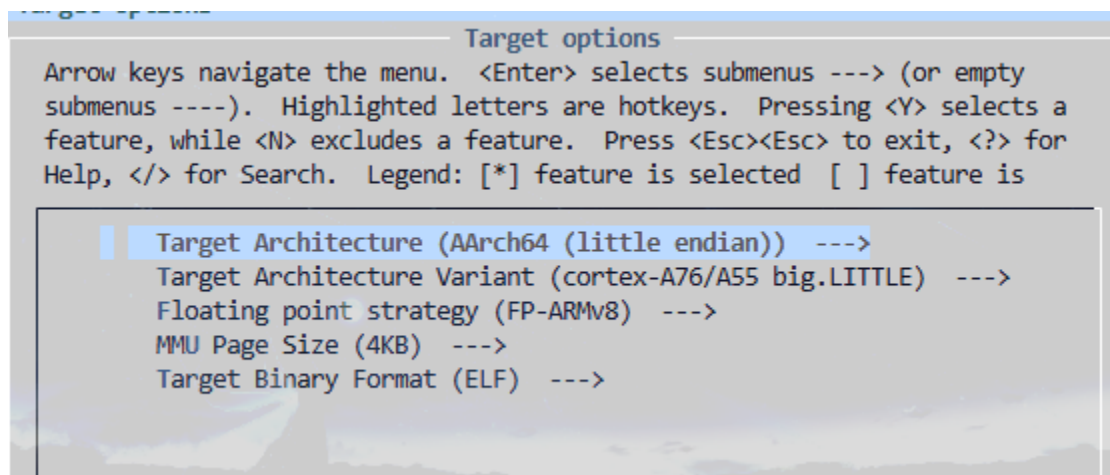
CROSS_COMPILE=这个参数按照实际情况填写。

下面说一下每个配置的情况。

1. 配置情况

执行 `./mk.sh menuconfig` 操作配置文件。

1.1 Target options



1.2 Toolchain

```
Toolchain
Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted
letters are hotkeys. Pressing <Y> selects a feature, while <N> excludes a feature. Press <Esc><Esc>
to exit, <?> for Help, </> for Search. Legend: [*] feature is selected [ ] feature is excluded

Toolchain type (External toolchain) --->
*** Toolchain External Options ***
Toolchain (Custom toolchain) --->
Toolchain origin (Pre-installed toolchain) --->
(/home/dev/3588_sdk/prebuilts/gcc/linux-x86/aarch64/gcc-arm-10.3-2021.07-x86_64-aarch64
(aarch64-none-linux-gnu) Toolchain prefix
External toolchain gcc version (10.x) --->
External toolchain kernel headers series (4.20.x) --->
External toolchain C library (glibc) --->
[*] Toolchain has SSP support?
[*] Toolchain has SSP strong support?
[ ] Toolchain has RPC support?
[*] Toolchain has C++ support?
[ ] Toolchain has D support?
[*] Toolchain has Fortran support?
[*] Toolchain has OpenMP support?
[ ] Copy gdb server to the Target
*** Host GDB Options ***
[ ] Build cross gdb for the host
*** Toolchain Generic Options ***
[ ] Copy gconv libraries
() Extra toolchain libraries to be copied to target
() Target Optimizations
() Target linker options
*** Bare metal toolchain ***
[ ] Build bare metal toolchains
```

我这里用的是 3588sdk 里面的交叉编译链，这部分的坑非常多，如果使用的是同一个交叉编译链请务必按照这样来配置。

1.3 Build options

这里基本按照文档来，但是要换源，回车进入 Mirrors and Download locations，修改成下面的网址：

```
BR2_BACKUP_SITE="http://sources.buildroot.net"
```

```
BR2_KERNEL_MIRROR="https://mirrors.aliyun.com/linux-kernel"
```

```
BR2_GNU_MIRROR="https://mirrors.aliyun.com/gnu"
```

```
BR2_LUAROCKS_MIRROR="https://luarocks.cn"
```

```
BR2_CPAN_MIRROR="https://mirrors.aliyun.com/CPAN"
```

1.4 System configuration

```
System configuration
Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are
hotkeys. Pressing <Y> selects a feature, while <N> excludes a feature. Press <Esc><Esc> to exit, <?> for Help, </>
for Search. Legend: [*] feature is selected [ ] feature is excluded

Root FS skeleton (default target skeleton) --->
(cat4) System hostname
(Welcome to Buildroot) System banner
  Passwords encoding (sha-256) --->
    Init system (BusyBox) --->
      /dev management (Dynamic using devtmpfs + mdev) --->
        (system/device_table.txt) Path to the permission tables
        [ ] support extended attributes in device tables
        [ ] Merged /usr
        [*] Enable root login with password
        (root) Root password
          /bin/sh (busybox' default shell) --->
            [*] Run a getty (login prompt) after boot --->
            [*] remount root filesystem read-write during boot
            ( ) Network interface to configure through DHCP
            (/bin:/sbin:/usr/bin:/usr/sbin) Set the system's default PATH
            [*] Purge unwanted locales
            (C en_US) Locales to keep
            ( ) Generate locale data
            [ ] Enable Native Language Support (NLS)
            [ ] Install timezone info
            ( ) Path to the users tables
            ( ) Root filesystem overlay directories
            ( ) Custom scripts to run before commencing the build
            ( ) Custom scripts to run before creating filesystem images
            ( ) Custom scripts to run inside the fakeroot environment
            ( ) Custom scripts to run after creating filesystem images
```

这里也基本按照文档来，只有部分按需修改。

点进 Run a getty (login prompt) after boot 这个配置，调整 TTY port 为 ttyFIQ0，如图：

```
/home/dev/self_linux/buildroot-2025.11/.config - Buildroot 2025.11 Configuration
→ System configuration → Run a getty (login prompt) after boot
  Run a getty (login prompt) after boot
  Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus
  ----). Highlighted letters are hotkeys. Pressing <Y> selects a feature, while
  <N> excludes a feature. Press <Esc><Esc> to exit, <?> for Help, </> for Search.
  Legend: [*] feature is selected [ ] feature is excluded

  --- Run a getty (login prompt) after boot
  (ttyFIQ0) TTY port
    Baudrate (keep kernel default) --->
  (vt100) TERM environment variable
  ( ) other options to pass to getty
```

1.5 Filesystem images

这里也是参考文档改成 ext4 文件系统，但是 additional mke2fs options 这个配置，要加上 -O ^metadata_csum_seed -O ^FEATURE_C12，编译出的 ext4 文件系统要去掉这两个特性。

```
Filesystem images
Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenu ----). Highlighted letters are
hotkeys. Pressing <Y> selects a feature, while <N> excludes a feature. Press <Esc><Esc> to exit, <?> for Help, </>
for Search. Legend: [*] feature is selected [ ] feature is excluded

[ ] btrfs root filesystem
[ ] cloop root filesystem for the target device
[ ] cpio the root filesystem (for use as an initial RAM filesystem)
[ ] cramfs root filesystem
[ ] erofs root filesystem
[*] ext2/3/4 root filesystem
    ext2/3/4 variant (ext4) --->
(rootfs) filesystem label
(256M) exact size
(0) exact number of inodes (leave at 0 for auto calculation)
(256) inode size
(5) reserved blocks percentage
(-0 ^64bit -0 ^metadata_csum_seed -0 ^FEATURE_C12) additional mke2fs options
    Compression method (no compression) --->
[ ] f2fs root filesystem
    *** initramfs needs a Linux kernel to be built ***
[ ] jffs2 root filesystem
[ ] oci image
[ ] squashfs root filesystem
[*] tar the root filesystem
    Compression method (no compression) --->
() other random options to pass to tar
[ ] ubi image containing an ubifs root filesystem
[ ] ubifs root filesystem
[ ] yaffs2 root filesystem
```

1.6 Kernel 和 uboot

因为我们只编译文件系统，所以这两个我们都不选中。

1.7 Target packages

这里就纯粹按需选择了，不做叙述。

配置完成后保存退出，执行./mk.sh，等待编译完成就好，生成的文件系统在 output/image 目录。

烧录进去之后需要自动挂载/boot 目录，不然设备树是错误的，也不能用设备树插件功能。

vi /etc/fstab

在里面加上/dev/mmcblk0p2 /boot auto defaults 0 2

需要开机初始化网络，修改/etc/network/interfaces，在 interfaces 文件添加 eth0 网卡配置：

auto eth0

iface eth0 inet dhcp